Recap Checkpoint 6 — Full Year 12 Cumulative (Pre-Mock): 1.1–1.5 + 2.1–2.3

Name: ____ Date: _ ***Mark:*** ___ / 40

OCR H446 cumulative recap — mixes every topic taught so far. Show all working.

Questions

Q1 [3] (AO1). *(1.1 — Processors)* State **three** factors that affect the **performance** of a processor, and for each give a brief explanation of its effect.

Q2 [4] (AO2). *(1.4 — Number bases)* Show all working.

(a) Convert denary **181** to **8-bit binary**. **[1]**

(b) Convert **181** to **hexadecimal**. **[2]**

(c) Convert binary `1010 0110` to denary. **[1]**

Q3 [4] (AO2). *(1.4 — Two's complement)* Using **8-bit two's complement**, calculate **−40 + 25**.

(a) Show 40 and 25 in 8-bit binary. **[1]**

(b) Form the two's complement representation of −40. **[1]**

(c) Perform the addition and give the final **denary** result. **[2]**

Q4 [3] (AO2). *(1.4 — Boolean algebra / floating point)*

(a) State the result of the floating-point benefit of **normalisation** (why numbers are normalised). **[1]**

(b) Simplify **A + A·B** and name the law used. **[2]**

Q5 [4] (AO1). *(1.3 — Networks)*

(a) Explain the difference between **circuit switching** and **packet switching**. **[2]**

(b) State **two** items of data added to a packet's **header**. **[2]**

Q6 [3] (AO1/AO3). *(1.5 — Legal/ethical)* Self-driving cars must sometimes "decide" how to act in an unavoidable collision.

Discuss **one ethical** issue this raises and identify **who** might be held **legally responsible** in a crash. **[3]**

Q7 [4] (AO2). *(2.1 / 2.2 — Computational thinking + programming)*

(a) State the difference between a **procedure** and a **function**. **[1]**

(b) Explain the difference between **passing a parameter by value** and **by reference**. **[2]**

(c) State **one** advantage of using **local** variables over **global** variables. **[1]**

Q8 [5] (AO2). *(2.3 — Algorithm trace)* A list contains `[7, 2, 9, 4, 1]`. Carry out a **bubble sort**, showing the list after **each complete pass**. State how many passes are needed before the list is sorted, and give the **worst-case** time complexity of bubble sort.

Q9 [4] (AO1/AO2). *(2.3 — Big O reasoning)*

(a) State the worst-case Big O of: insertion sort, merge sort, and a binary search. **[3]**

(b) Explain why merge sort is preferable to insertion sort for very large datasets, referencing growth rates. **[1]**

Q10 [6] (AO3). *(2.3 — Graphs, applied)* Using **Dijkstra's algorithm**, find the shortest **distance** and shortest **path** from **A** to **E** in the weighted, undirected graph below. Show a table of tentative distances and the order in which nodes are visited.

```
Edges (undirected, weighted):
A-B = 2,  A-C = 5,  B-C = 1,  B-D = 7,  C-D = 3,  C-E = 8,  D-E = 2
```

Q11 [4 + (rest)] (AO3). *(Synoptic extended response)* A hospital is replacing a paper appointment system with networked software that stores patient data, lets staff search records, and sends automated reminders. In an extended response, discuss the design considering: **(i)** how **abstraction and decomposition** help manage the build, **(ii)** an appropriate **data structure or search algorithm** for fast record lookup (justify with time complexity), and **(iii) one legal** consideration for storing patient data. **[6]**